

KALEVALA, Russian Federation

WMO: 224080

Lat: **65.217N**Lon: **31.150E**Elev: **120**StdP: **99.89**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.8	-29.2	-35.2	0.1	-32.8	-31.3	0.2	-29.2	7.5	-6.1	6.6	-5.0	0.4	340	0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	25.6	18.0	23.5	17.0	21.8	16.0	19.0	24.0	17.9	22.3	16.7	20.8	2.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.5	21.0	16.0	11.5	20.3	14.8	10.6	19.1	54.5	24.1	50.9	22.3	47.3	20.9	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.8	5.9	5.1	-35.7	27.7	4.1	2.1	-38.6	29.2	-41.0	30.4	-43.3	31.6	-46.2	33.1		
			-35.7	20.4	4.0	1.7	-38.6	21.6	-41.0	22.6	-43.3	23.6	-46.2	24.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.4	-11.5	-11.4	-6.3	0.0	6.4	12.3	15.8	13.1	8.1	2.1	-3.9	-8.3
	DBStd	10.87	8.38	8.35	5.80	4.50	4.68	4.01	3.52	3.41	3.56	4.12	5.74	7.78
	HDD10.0	3543	666	598	507	301	133	21	2	9	78	244	418	567
	HDD18.3	6191	924	831	765	551	372	186	94	166	307	502	668	826
	CDD10.0	417	0	0	0	0	20	89	182	105	21	1	0	0
	CDD18.3	22	0	0	0	0	0	4	15	3	0	0	0	0
	CDH23.3	185	0	0	0	0	8	44	105	28	0	0	0	0
	CDH26.7	22	0	0	0	0	1	4	14	3	0	0	0	0
Wind	WSAvg	2.0	1.9	2.0	2.3	2.0	2.2	2.1	1.7	1.6	1.9	2.2	2.1	2.0
Precipitation	PrecAvg	532	30	25	24	27	39	61	68	74	63	46	45	31
	PrecMax	718	71	65	53	83	77	110	198	123	105	89	90	54
	PrecMin	414	7	5	3	3	4	29	17	29	25	12	3	7
	PrecStd	73	14	15	13	17	19	25	35	25	19	21	18	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.6	3.8	6.9	15.0	24.4	27.2	28.6	26.2	19.8	12.4	7.9	4.0
		MCWB	1.7	1.9	3.1	9.3	15.0	18.2	19.7	19.0	14.6	10.5	6.5	2.8
	2%	DB	1.1	1.7	4.4	10.7	20.7	24.4	26.1	23.6	17.3	10.3	5.7	2.4
		MCWB	0.5	0.8	1.5	6.1	13.0	16.9	18.6	17.5	13.0	8.9	4.9	1.6
	5%	DB	0.2	0.6	2.6	8.5	17.3	21.9	24.2	21.5	15.4	9.1	3.8	1.4
		MCWB	-0.2	-0.1	0.6	4.4	10.9	15.5	17.8	16.3	12.1	8.1	3.1	0.8
	10%	DB	-1.1	-0.7	1.2	6.5	14.8	19.7	22.4	19.5	13.9	7.7	2.3	0.4
		MCWB	-1.6	-1.4	-0.2	3.2	9.5	14.1	16.6	15.2	11.5	6.7	1.8	0.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.9	2.2	3.5	9.0	16.2	19.4	21.4	19.9	15.3	10.7	6.9	3.0
		MCDB	2.7	3.2	6.1	13.8	23.3	25.3	26.3	24.2	18.6	11.7	7.3	3.3
	2%	WB	0.6	1.0	1.9	6.5	13.6	17.9	19.5	18.0	13.8	9.4	4.9	1.8
		MCDB	1.0	1.7	3.7	10.2	19.4	23.2	24.4	22.6	16.2	10.1	5.6	2.2
	5%	WB	-0.2	0.1	0.9	4.8	11.5	16.4	18.5	16.8	12.7	8.1	3.2	0.8
		MCDB	0.2	0.6	2.1	7.8	16.6	21.0	22.9	20.6	14.8	8.9	3.7	1.3
	10%	WB	-1.5	-1.4	0.0	3.5	9.9	14.5	17.4	15.6	11.7	6.8	1.9	0.0
		MCDB	-1.1	-0.7	1.1	6.1	14.5	18.7	21.3	18.9	13.5	7.6	2.3	0.4
Mean Daily Temperature Range	5% DB	MDBR	7.3	7.9	9.1	8.8	9.2	9.2	8.9	8.8	7.3	4.5	4.8	6.1
		MCDBR	5.5	5.8	8.6	11.9	14.2	12.8	11.7	12.1	10.1	5.3	4.1	4.2
		MCWBR	5.4	5.4	6.9	7.7	8.1	6.7	5.8	6.8	6.4	4.5	3.9	4.0
	5% WB	MCDBR	5.3	5.7	7.0	10.8	13.1	11.4	10.4	10.6	8.7	5.0	3.8	4.1
		MCWBR	5.3	5.4	5.8	7.2	8.0	6.6	5.7	6.3	6.0	4.5	3.8	4.1
Clear-Sky Solar Irradiance	taub	0.204	0.272	0.303	0.336	0.347	0.343	0.368	0.344	0.320	0.295	0.352	0.202	
	taud	1.932	2.100	2.162	2.292	2.448	2.488	2.442	2.482	2.529	2.374	2.640	1.957	
	Ebn at Noon	284	598	760	830	861	872	834	822	766	594	278	68	
	Edh at Noon	22	59	88	100	95	93	95	83	64	46	21	7	
All-Sky Solar Radiation	RadAvg	0.12	0.67	2.22	3.87	4.60	5.12	4.78	3.50	1.95	0.68	0.15	0.03	
	RadStd	0.01	0.13	0.20	0.32	0.49	0.35	0.51	0.33	0.21	0.10	0.03	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.61	N/A	N/A	N/A	N/A	N/A	N/A	-242	N/A	N/A
(47) Regional (16 neighbors)		+0.57	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page